

Analytical Solutions for One-Dimensional Water Flow in Vegetated Layered Soil

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Abstract: Vegetation reduces soil pore-water pressure (PWP) through root water uptake. There are limited studies about the effects of vegetation on PWP distributions in layered soil, such as multilayered earthen landfill cover. New analytical solutions for vertical water flow in vegetated layered soil at steady state are derived considering various root architectures, including exponential, parabolic, triangular, and uniform roots. The solutions are verified through (1) hypothetical vegetated layered soil using numerical seepage analyses and (2) a soil column test investigating PWP distributions in vegetated layered soil. The new solutions are capable of determining PWP distributions for both inside and outside root zone in vegetated layered soil. These solutions provide a relatively simple and reliable approach for estimation of PWP distributions in vegetated layered soil. DOI: [10.1061/\(ASCE\)GM.1943-5622.0001343](https://doi.org/10.1061/(ASCE)GM.1943-5622.0001343). © 2018 American Society of Civil Engineers.

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Introduction

Vegetation provides an environment-friendly method for reducing water infiltration and enhancing slope stability through root water uptake in civil infrastructures, such as embankments and landfill covers. Vegetation reduces the pore-water pressure (PWP) through root water uptake. Thereafter, shear strength of soil increases but water permeability decreases (Ng and Menzies 2007; Liu et al. 2016). Hence, slope stability could be improved, and water percolation in landfill cover might be reduced (Nyambayo and Potts 2010; Ng et al. 2013).

Layered soil is frequently found in civil infrastructures; for instance, landfill cover or slope commonly consists of multiple soil layers (Feng et al. 2017a, b; Rahardjo et al. 2012; Ng et al. 2016). Based on design guidelines (Bonaparte et al. 2004), the top layer of landfill cover is the vegetated layer. Vegetation not only helps protect the underlying soil layers against wind/water erosion, but it also recovers the water storage capacity of landfill cover through root water uptake during the drying period (Feng et al. 2017a). Absorption of soil water content by vegetation affects PWP distributions in soil, which in turn affects total head distributions and, hence, unsaturated seepage in soil. For layered soil, continuity of flux and equality of PWP should be satisfied at the interface of any two consecutive layers (Yeh 1989). Underneath the vegetated layer,

different types of landfill covers with layered soils are commonly used. For instance, the traditional landfill cover with capillary barrier effect (CCBE), consisting of a fine-particle soil (i.e., clay/silt) layer overlying a coarse-particle soil (i.e., gravel/sand), is frequently used in landfill sites to minimize water percolation. Ng et al. (2016) proposed a new three-layered landfill cover by adding a clay layer below the CCBE. Both soil column test and numerical analysis show that the new three-layer landfill works well in humid regions, whereas the CCBE is only suitable for arid/semiarid regions (Ng et al. 2016). Understanding the response of soil hydrology in vegetated layered soil is necessary for engineers to better use vegetation for enhancing slope stability and reducing water percolation.

Analytical and numerical methods have been developed for capturing PWP distributions in single-layered vegetated soil (Raats 1974; Nyambayo and Potts 2010; Prasad 1988; Zaradny et al. 1978; Yuan and Lu 2005). A sink term is usually used to simulate the root water uptake and is incorporated into the governing equation of unsaturated seepage [i.e., the Richard's equation; Richards (1931)]. Numerical simulations were conducted by Prasad (1988) and Zaradny et al. (1978) to investigate one-dimensional (1D) water flow in vegetated ground, considering uniform and triangular roots, respectively. Analytical solutions are suitable tools to verify numerical simulations, which usually suffer from convergence. A steady-state analytical solution for 1D unsaturated seepage in flat ground was proposed by Raats (1974), considering exponential root architecture. Only the variations in water content distributions due to root water uptake, rather than PWP distributions, were examined by Raats (1974), Prasad (1988), and Zaradny et al. (1978). Recently, analytical solutions of PWP distributions in a single-layer vegetated soil were developed by Ng et al. (2015), considering different root architectures. It was found that root architecture affects root water uptake distributions, leading to different PWP distributions (Ng et al. 2015). However, due to the assumption of a single soil layer made in the previously mentioned valuable research, root architectures are not applicable to layered soil, such as layered earthen landfill cover (Ng et al. 2016; Feng et al. 2017a, b). Although analytical solutions are available for 1D water infiltration in layered soils (Wu et al. 2012; Srivastava and Yeh 1991; Zhan and Ng 2004; Zhan et al. 2013), the effects of root water uptake are generally ignored. As far as the authors are aware, analytical solutions of PWP distributions in vegetated layered soil have not been available. Compared with

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numerical simulations, analytical solutions usually have the advantages of simplicity and explicitness (Wu et al. 2012). Thus, it would be expected that such analytical solutions could provide a simple and robust method for studying effects of vegetation on soil hydrology of civil infrastructures, in which layered soil are frequently found.

This research aims to derive analytical solutions of PWP distributions in vegetated layered soil. The analytical solutions are verified through numerical simulations and a selected soil column test with vegetation. The analytical solutions can provide a simple but robust method for studying the variations in PWP due to root water uptake in vegetated layered soil, such as multilayer earthen landfill cover.

Theoretical Formulations

Governing Equation and Boundary Conditions

Fig. 1 presents multilayer vegetated soil, which contains j ($j \geq 2$) layers of soil. The top layer is the vegetated layer. The governing equation describing 1D unsaturated seepage in the vegetated soil could be expressed as follows (Yuan and Lu 2005):

$$\frac{\partial^2 k_i}{\partial z_i^2} + \alpha_i \frac{\partial k_i}{\partial z_i} - \alpha_i S(z_i) H(z_i) = 0 \quad (1)$$

where k_i = water permeability of i th ($1 \leq i \leq j$) soil layer; z_i = the coordinate perpendicular to the soil surface with the origin point at the bottom of the i th layer as presented in Fig. 1; α_i = desaturation coefficient of the i th soil layer [describing the desaturation rate due

to increase in negative PWP; Zhan and Ng (2004)]; L_1 = depth outside root zone (Fig. 1); and $S(z_i)$ = root water uptake, which considers different root architectures (Fig. 1) and is defined as follows (Liu et al. 2016; Feng et al. 2017a):

$$S(z_i) = \begin{cases} \frac{T_p}{L_2} \\ \frac{2T_p}{L_2} \left(\frac{L_2 - z_i}{L_2} \right) \\ T_p \left[\frac{\exp(z_i - L_1) - 1}{\exp(L_2) - L_2 - 1} \right] \\ \frac{2T_p}{L_2} \left[\frac{3 \left((z_i - L_1)L_2 - (z_i - L_1)^2 \right)}{L_2} \right] \end{cases} \quad (2)$$

where T_p = transpiration rate; and L_2 = root depth (Fig. 1). Note that in Eq. (2) it is assumed that the distribution of root water uptake is proportional to root architecture (Ng et al. 2015). For a fair comparison, the total root area for the different root architectures is identical [i.e., the same root water uptake for a given transpiration rate (T_p)].

$H(z_i)$ is defined as

$$H(z_i) = \begin{cases} 1 & i = 1 \text{ inside root zone (i.e., 1st layer)} \\ 0 & i \neq 1 \text{ outside root zone} \end{cases} \quad (3)$$

Eq. (1) physically models that water flow driven by the PWP gradient ($\partial^2 k_i / \partial z_i^2$), and gravity ($\alpha_i \partial k_i / \partial z_i$) is balanced by root water uptake [$\alpha_i S(z_i) H(z_i)$] at steady state.

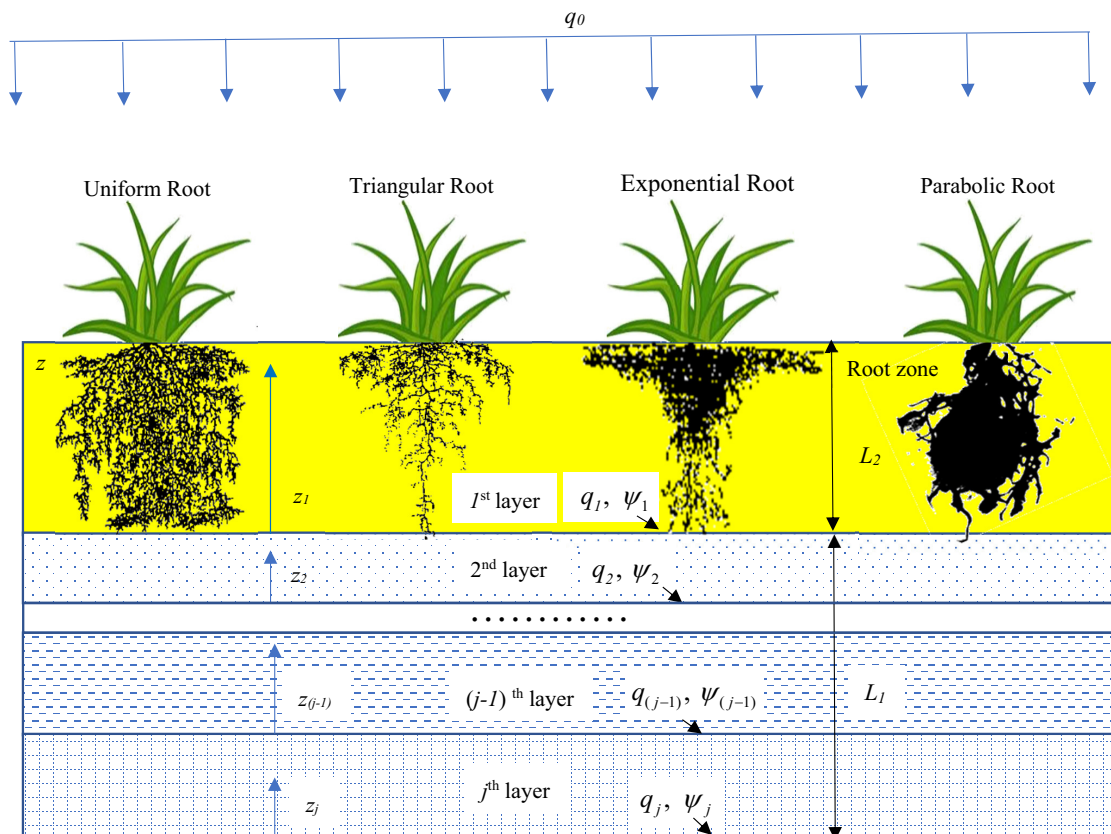


Fig. 1. Schematic diagram of vegetated layered soil with different root architectures.

The formulations describing unsaturated water permeability function and volumetric water content of soil are given as follows (Gardner 1958):

$$k_i = k_{si} \exp(\alpha_i \psi) \quad (4)$$

$$\theta_{wi} = \theta_{ri} + (\theta_{si} - \theta_{ri}) \exp(\alpha_i \psi) \quad (5)$$

where k_{si} = saturated water permeability of the i th soil layer; ψ = PWP head; θ_{ri} = residual volumetric water content of the i th soil layer; and θ_{si} = saturated volumetric water content of the i th soil layer. Although the hydrological behavior of soil near saturation may not be described well by Eqs. (4) and (5), both equations have been adopted for deriving analytical solutions for unsaturated seepage due to mathematical convenience (Gardner 1958; Yuan and Lu 2005; Zhan et al. 2013; Srivastava and Yeh 1991). Because only the drying condition is considered in this study, only the drying path of the soil water characteristic curve [Eq. (5)] is adopted; hence, the hysteresis effect is neglected in this study. It has been revealed that soil hydraulic properties could be affected by roots (Ni et al. 2018; Khalilzad et al. 2014; Jotisankasa and Sirirattanachai 2017; Leung et al. 2015). However, the reported results are contradictory because some studies found an increase in saturated permeability, whereas some reported a decrease. Hence, the influence of vegetation on soil hydraulic properties is ignored in this study. The spatial variations of soil hydraulic properties and plant characteristics have been recognized (Das et al. 2018; Yeh 1989; Gadi et al. 2017); however, they are not in the scope of this study. Statistical/probabilistic analysis of results (Das et al. 2018; Garg et al. 2017) could be considered for future research.

A constant PWP is specified at the bottom boundary as follows:

$$k_j|_{z_j=0} = k_{sj} \exp(\alpha_j \psi_j) \quad (6)$$

where ψ_j = PWP head at the bottom boundary of the j th layer (Fig. 1).

The top boundary is defined as a constant flux boundary (i.e., evaporation or rainfall flux), which is given as follows:

$$\left[\frac{\partial k_1}{\partial z_1} + \alpha_1 k_1 \right]_{z_1=L_2} = -\alpha_1 q_0 \quad (7)$$

where q_0 = surface flux (i.e., positive value for evaporation, while negative value for rainfall).

Derivation of Analytical Solutions

At steady state, the PWP head and water flux of any two successive layers are the same at the interface. Based on water mass balance, water flux at the interface can be determined (see subsequent detailed information). For vegetated layered soils, the solution can be first derived for the bottom layer (Fig. 1, j th layer), which has a bottom boundary of a constant PWP head ψ_j [Eq. (6)] and a top boundary of water flux of q_{j-1} [Eq. (7)]. Thereafter, the PWP head at the top of the j th layer (Fig. 1, ψ_{j-1}) can be calculated. Then

PWP distributions in the overlying layer [Fig. 1, $(j-1)$ th layer] can be evaluated by using ψ_{j-1} as the prescribed head at the bottom boundary and q_{j-2} (Fig. 1) as the water flux at the top boundary. Using this procedure recursively, PWP distributions in each soil layer can be calculated. The detailed information is given as follows.

Outside Root Zone

According to water mass balance, the vertical water flux outside root zone [q_i ($i \neq 1$)] is the same at steady state, equaling to the sum of water flux (Fig. 1, q_0) and total root water uptake amount (T_p). Hence, q_i ($i \neq 1$) can be described as follows:

$$q_i = q_0 + T_p \quad (i \neq 1) \quad (8)$$

Based on Darcy's law, vertical water flux outside root zone is given as follows:

$$q_i = -k_i \left(\frac{d\psi_i}{dz_i} + 1 \right) \quad (i \neq 1) \quad (9)$$

By integrating Eq. (9) along the thickness of the i th soil layer, the PWP distributions in the i th layer (u_i) can be determined as follows:

$$u_i = 10\alpha_i^{-1} \ln \left\{ \exp[\alpha_i(\psi_i - z_i)] + \frac{1}{k_{si}} (q_0 + T_p) [\exp(-\alpha_i z_i) - 1] \right\} \quad (10)$$

In Eq. (10), the first term, $\exp[\alpha_i(\psi_i - z_i)]$, represents the effects of PWP specified at the bottom boundary of the i th soil, whereas the second term describes the combined influences of root water uptake and rainfall/evaporation.

For layered soil, Eq. (10) can be applied to the bottom layer (i.e., Fig. 1, j th layer) first to obtain the PWP at the interface between the j th layer and the overlying layer [i.e., Fig. 1, $(j-1)$ th layer]. Because the PWP in the j th layer equals that in the $(j-1)$ th layer at the interface, the PWP distribution in the $(j-1)$ th layer can be evaluated by using Eq. (10) and the PWP at the interface as the prescribed value (i.e., Fig. 1, ψ_{j-1}) at the bottom boundary of the $(j-1)$ th layer. By using this procedure recursively, the PWP profiles in the layered soil outside root zone can be determined.

Inside Root Zone

The PWP (i.e., Fig. 1, ψ_1) at the bottom of the vegetation layer (i.e., Fig. 1, 1st layer) equals that at the top of the 2nd layer (outside root zone). The PWP (i.e., Fig. 1, ψ_1) at the top of the 2nd layer is determined by Eq. (10). Hence, the PWP distributions in the vegetation layer can be calculated by using the steady-state solutions of PWP in the single-layer vegetated soil derived by Ng et al. (2015). Correspondingly, for the top vegetation layer, the lower boundary is specified as the prescribed head (ψ_1), whereas evaporation/rainfall flux is considered at the upper boundary. Thereafter, the steady-state solution of PWP in the vegetation layer is derived as follows:

$$u_w = 10\alpha_1^{-1} \ln \left\{ \exp[\alpha_1(\psi_1 - z_1)] + \frac{q_0}{k_{s1}} [\exp(-\alpha_1 z_1) - 1] + \frac{\alpha_1}{k_{s1}} \int_0^{L_2} G(z_1, x) S(x) H(z_1) dx \right\} \quad (11)$$

where the definition of $G(z_1, x)$ is given as follows:

$$G(z_1, x) = \begin{cases} \exp(-\alpha_1 z_1) [1 - \exp(\alpha_1 x)] / \alpha_1 & 0 \leq x \leq z_1 \leq L_2 \\ [\exp(-\alpha_1 z_1) - 1] / \alpha_1 & 0 \leq z_1 \leq x \leq L_2 \end{cases} \quad (12)$$

Eq. (11) models that PWP profiles in vegetated soil are affected by (1) PWP profiles at hydrostatic state (the first term), (2) effects of rainfall/evaporation on PWP distributions (the second term), and (3) effects of vegetation (the third term).

After substituting Eq. (2) and Eq. (12) into Eq. (11), the analytical solutions of steady-state PWP distributions considering different root architectures within root zone are derived as follows:

$$u_i = 10\alpha_1^{-1} \ln k_r \quad (13)$$

where k_r = relative water permeability.

For uniform root architecture

$$k_r = A + \frac{1}{k_{s1}} \frac{T_p}{L_2} \left\{ [\exp(-a_1 z_1) - 1] (L_2 - z_1) + \exp(-a_1 z_1) \left[z_1 - a_1^{-1} \exp(a_1 z_1) + a_1^{-1} \right] \right\} \quad (14)$$

where $A = \exp[a_1(\psi_1 - z_1)] + q_s [\exp(-a_1 z_1) - 1] / k_{s1}$

For triangular root architecture

$$k_r = A + \frac{a_1}{k_{s1}} \frac{2T_p}{L_2^2} \left\{ \frac{1}{a_1} [\exp(-a_1 z_1) - 1] \left(\frac{L_2^2 - z_1^2}{2} \right) + \frac{\exp(-a_1 z_1)}{a_1^3} - \frac{a_1 z_1 - 1}{a_1} \right\} + \frac{z_1^2 \exp(-a_1 z_1)}{2a_1} \quad (15)$$

For exponential root architecture

$$k_r = A + \frac{a_1 T_p}{k_{s1} (\exp(L_2) - L_2 - 1)} \left\{ \frac{\exp(-a_1 z_1) (\exp(z_1) - 1)}{a_1} - \frac{z_1 \exp(-a_1 z_1)}{a_1} - \frac{\exp(-a_1 z_1) (\exp((a_1 + 1)z_1) - 1)}{a_1 (a_1 + 1)} + \frac{\exp(-a_1 z_1) (\exp(a_1 z_1) - 1)}{a_1^2} \right. \\ \left. + \frac{\exp(-a_1 z_1) - 1}{a_1} (\exp(L_2) - \exp(z_1) - L_2 + z_1) \right\} \quad (16)$$

For parabolic root architecture

$$k_r = A + \frac{1}{k_{s1}} \frac{6a_1 T_p}{L_2^3} \left\{ \frac{z_1^2 L_2 \exp(-a_1 z_1)}{2a_1} - \frac{z_1^3}{3a_1} \exp(-a_1 z_1) - L_2 \exp(-a_1 z_1) \left[\frac{z_1 \exp(a_1 z_1)}{a_1^2} - \frac{\exp(a_1 z_1) - 1}{a_1^3} \right] \right. \\ \left. + \frac{\exp(-a_1 z_1)}{a_1} \left[\frac{z_1^2 \exp(a_1 z_1)}{a_1} - \frac{2z_1 \exp(a_1 z_1)}{a_1^2} + \frac{2 \exp(a_1 z_1)}{a_1^3} - \frac{2}{a_1^3} \right] \right. \\ \left. + \frac{\exp(-a_1 z_1) - 1}{a_1} \left[\frac{(L_2^2 - z_1^2) L_2}{2} - \frac{L_2^3 - z_1^3}{3} \right] \right\} \quad (17)$$

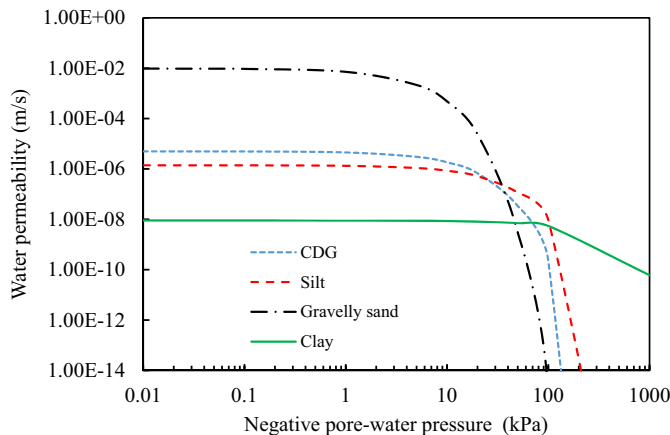


Fig. 2. Water permeability functions of different soils used in the analytical solutions.

Verification of Analytical Solutions

Phase 1: Verified by Numerical Simulation

The new solutions of PWP in the multilayer vegetated soils are verified by 1D numerical simulation. The mathematics module of COMSOL, which is a commercial software developed based on FEM, is adopted (COMSOL 2015). This mathematics module provides a flexible user-defined interface for specifying and solving a set of partial differential equations for 1D problems. The governing equation of 1D water flow in the vegetated soil [Eq. (1)] is custom built in COMSOL. The sink term $S(z_i)$ as defined by Eq. (3) is adopted to model root water uptake. The water permeability function is defined by Eq. (4), whereas the soil water characteristic curve is considered by Eq. (5). Combined with the boundary conditions [Eqs. (6) and (7)], the proposed theoretical model defined by Eqs. (1)–(7) are numerically solved by the nonlinear solver provided by COMSOL. The bottom boundary and the top boundary of the numerical model are specified as a water table and zero flux boundary

(i.e., ignoring surface evaporation), respectively. Two soil layers are considered in the model, including a top 1-m layer of completely decomposed granite (CDG) and a bottom 1-m layer of clay. The permeability functions of CDG and clay are provided in Fig. (2), which also provide the water permeability functions of gravelly sand and silt that are used to explore PWP distributions in a layered earthen landfill cover (see subsequent detailed information). Root depth is 0.5 m. Four root architectures described by Eq. (2) are considered. Based on Leung and Ng (2013), the transpiration rate is 2.5 mm/day. Steady-state simulation is conducted.

Fig. 3(a) illustrates that the analytical and numerical PWP profiles for both parabolic and uniform root architectures are

consistent. As expected, the vegetated soil has larger negative PWP than bare soil (hydrostatic pressure) due to root water uptake. The parabolic and uniform roots have nearly identical PWP profiles. The negative PWP in the bottom clay layer increases from 0 to 46 kPa from the bottom to the top, whereas an increase in negative PWP of only 20 kPa can be observed for the top CDG layer; that is, the larger hydraulic gradient occurs in the bottom soil, which is outside root zone. This is different with the calculated PWP response in a vegetated single-layer soil (Ng et al. 2015), which shows that the larger hydraulic gradient occurs in the zone within root depth at drying steady state. This is because in the two-layered soil the water permeability of the bottom clay layer is smaller than that of the top

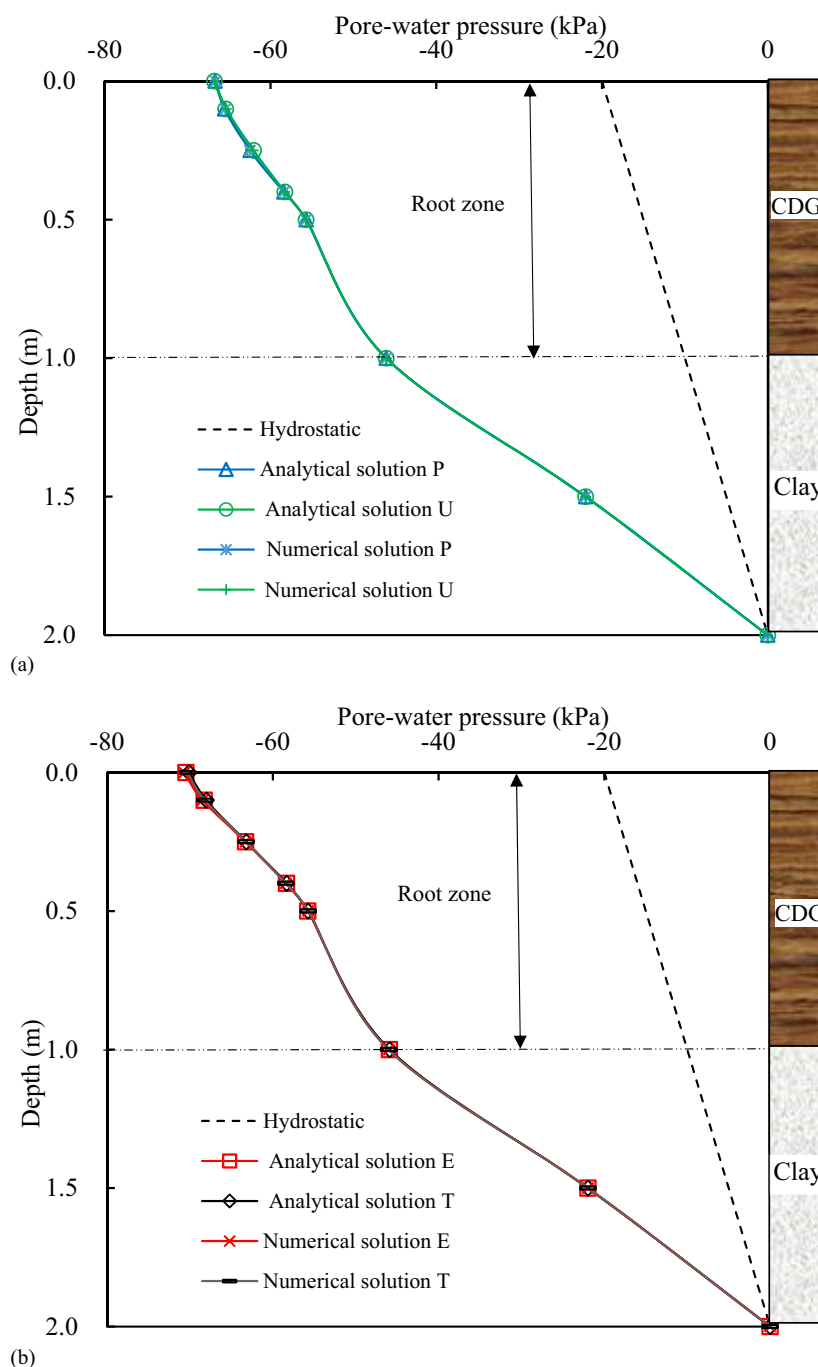


Fig. 3. Verification of the analytical solutions for calculating PWP in vegetated layered soil: (a) parabolic (P) and uniform (U) root architectures; and (b) exponential (E) and triangular (T) root architectures.

CDG layer for a negative PWP lower than 100 kPa (Fig. 2). Hence, for a given root water uptake, larger hydraulic gradient is induced in the clay layer based on Darcy's law. These results illustrate that the water permeability of the soil outside root zone affects PWP response within root zone. As the water permeability of soil outside root zone decreases, a larger hydraulic gradient induced by root water uptake can be observed in soil outside root zone compared with that within root zone. These results suggest that soil with low water permeability, such as fine-grained soil (i.e., clay) or soil with a high degree of compaction, could be used outside root zone to amplify the beneficial effects of vegetation on PWP reduction and hence slope stabilization.

Fig. 3(b) illustrates similar PWP distributions for exponential and triangular rooted soils. Compared with Fig. 3(a), exponential and triangular roots generated a larger negative PWP inside root zone. This is because more roots are available in exponential and triangular roots near the slope surface, leading to more root water uptake close to the slope surface and a larger negative PWP. In contrast, different root architectures have negligible effects on the PWP distributions outside root zone [Figs. 3(a and b)]. Ng et al. (2015) reported similar results, which show PWP distributions outside root zone are almost independent with root architecture. These results illustrate that exponential or triangular roots are more suitable for inducing larger negative PWP. Therefore, the beneficial effects of roots on shallow slope stabilization as well as mitigation of soil erosion could be amplified.

Phase 2: Verified by Selected Case Study

The soil column test performed by Liu (2017) is adopted. The soil column contains four layers of soil from the bottom to the top, namely a 400-mm clay layer, 200-mm gravelly sand layer, 400-mm silt layer, and 300-mm CDG layer. The bottom three layers constitute the new three-layer soil column proposed by Ng et al. (2016), whereas the top layer serves as the vegetation layer. After soil compaction, vetiver grass, which is tall perennial grass and commonly

found in most areas of Asia (i.e., southern part of China, Malaysia, and Thailand; Hau and Corlett 2003), was transplanted. The test was conducted with room temperatures of 23–25°C. Nine measurements were made at 100, 300, 500, 700, 900, 1,090, 1,160, 1,220, and 1,270 mm above the base of the soil column. At each measurement location, a small-tip tensiometer [measurement range: 0–90 kPa; accuracy: 1 kPa; Soilmoisture Equipment Corporation (2009)] was adopted to measure PWP.

The 1D analytical simulation is conducted following the reported height of the soil column tests. The measured evaporation rate of 0.5 mm/day is specified at the top boundary [i.e., $q_s = -0.5$ mm/day in Eq. (7)], whereas the bottom boundary is the measured PWP of 0 kPa [i.e., $\psi_1 = 0$ kPa in Eq. (6)]. The uniform root architecture with a root depth of 200 mm is adopted based on the measurements by Liu (2017). The measured transpiration rate T_p of 1.5 mm/day is used. Fig. 2 presents the hydraulic properties of the four soil types [i.e., CDG (silty sand), silt, gravelly sand, and clay].

Comparison between calculated and measured PWP distributions is provided in Fig. 4. Generally, the calculated results have a trend similar to that of the measurements. The vegetated soil has larger negative PWP than hydrostatic pressure. This is expected due to root water uptake in vegetated soil. The difference between negative PWP in vegetated soil and hydrostatic pressure decreases as depth increases. The influence zone of root water uptake is about six times root depth, which is consistent with the measured range of two to six times root depth in laboratory tests, which investigated the influence of vegetation on PWP distributions (Leung and Ng 2013; Ng et al. 2013). The largest hydraulic gradient occurs in the clay layer because clay has the lowest water permeability for the range of measured negative PWP (Fig. 2). Hence, the largest hydraulic gradient is induced to compensate for the amount of root water uptake, following Darcy's law. The calculated negative PWP is slightly overestimated, especially outside root zone. For the measured and calculated results, the maximum difference about 7 kPa is observed at the 0.8-m depth in the sand layer. This is probably because the exponential form of water permeability function

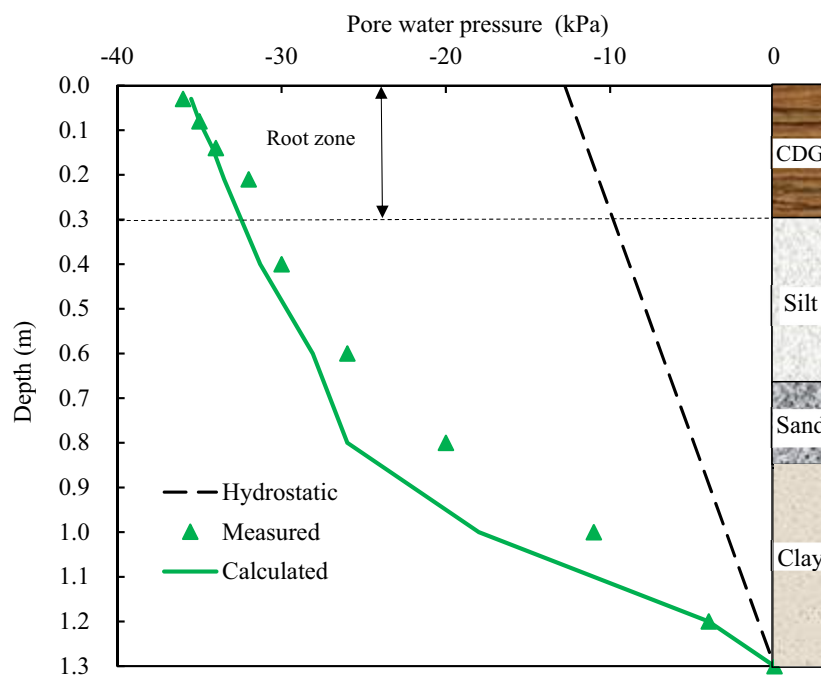


Fig. 4. Comparisons between measured and calculated PWP distributions in the vetiver-planted layered soil.

[Eq. (4)] may not be well applied for a wide range of PWP (Gardner 1958; Yeh 1989). The overestimation of the calculated negative PWP indicates the underestimation of unsaturated water permeability [Eq. (4)] for soils outside root zone. According to Darcy's law, a larger hydraulic gradient (i.e., a larger negative PWP) is needed for inducing a given water flux [Eq. (8)] outside root zone. These results illustrate that for the exponential form water permeability function [Eq. (4)], the selection of parameters for a given soil may depend on the range of the PWP.

Summary and Conclusions

This study derives analytical solutions of steady-state PWP distributions in vegetated layered soils. Four root architectures are considered, including parabolic, exponential, uniform, and triangular roots. The analytical solutions are verified by (1) a 1D numerical model simulating PWP distributions at steady state and (2) soil column tests investigating PWP distributions in vegetated layered soil. The new method can determine PWP distributions in vegetated layered soil reasonably. This method provides a relatively simple and reliable approach for the estimation of PWP distributions in vegetated layered soil, such as layered earthen landfill cover.

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Notation

The following symbols are used in this paper:

- $G(z_1, x)$ = green function;
- $H(z_i) = \begin{cases} 1 & i = 1 \text{ inside root zone (i.e., 1st layer)} \\ 0 & i \neq 1 \text{ outside root zone} \end{cases}$;
- j = number of soil layers;
- k_i = water permeability of the i th ($1 \leq i \leq j$) soil layer;
- k_r = relative water permeability;
- k_{si} = saturated water permeability of the i th soil layer;
- L_1 = depth outside root zone;
- L_2 = root depth;
- q_i ($i \neq 1$) = vertical water flux outside root zone for the i th ($i \neq 1$) layer;
- q_0 = surface flux;
- $S(z_i)$ = root water uptake;
- T_p = transpiration rate;
- u_i = PWP distributions in the i th layer;
- x = integration variable;
- z_i = coordinate perpendicular to the soil surface with origin point at the bottom of the i th layer;
- α_i = desaturation coefficient of the i th soil layer;
- θ_{ri} = residual volumetric water content of the i th soil layer;
- θ_{si} = saturated volumetric water content of the i th soil layer; and
- ψ_j = PWP head at the bottom boundary of the j th layer.

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